INCH-POUND
MIL-DTL-24643/50E
1 October 2009
SUPERSEDING
MIL-DTL-24643/50D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 1000 VOLTS, TYPE LSFNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation (see table I for thickness). Standard identification code shall be applied by Method 1 or Method 3. If identification Method 1 is used, the insulation color shall be black.
Fourth	-	Four conductors cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm well-rounded assembly.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/50	Type and size	AWG conductor size	Insulation thickness min. avg. (inch)	Cable jacket thickness min. avg. (inch)	Overall diameter	
					min. (inch)	max. (inch)
-01UN	LSFNW-3	16 (Class B)	0.030	0.040	0.413	0.447
-02UN	LSFNW-4	14 (Class B)	0.030	0.040	0.475	0.513
-03UN	LSFNW-9	10 (Class B)	0.030	0.040	0.583	0.630
-04UN	LSFNW-23	7 (Class B)	0.045	0.055	0.768	0.830
-05UN	LSFNW-50	3 (Class C)	0.062	0.055	1.016	1.050
-06UN	LSFNW-75	1 (Class C)	0.062	0.065	1.206	1.240
-07UN	LSFNW-100	0 (Class D)	0.062	0.065	1.324	1.385
-08UN	LSFNW-150	000 (Class D)	0.062	0.065	1.591	1.625
-09UN	LSFNW-200	0000 (Class D)	0.072	0.065	1.786	1.820

TABLE II. Electrical details.

Type and size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand min. (volts)	Accelerated service loading current (amperes)
LSFNW-3	4.3	6000	4000	N/A
LSFNW-4	2.68	5000	4000	N/A
LSFNW-9	1.08	3800	4000	N/A
LSFNW-23	0.543	3700	5500	115
LSFNW-50	0.210	3000	6000	215
LSFNW-75	0.134	3000	6000	285
LSFNW-100	0.106	2500	6000	335
LSFNW-150	0.0674	2500	6000	440
LSFNW-200	0.0536	2000	6500	525

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.) — See table II

Voltage withstand (volts, root mean square, min.)

Conductor to conductor — See table II

Insulation resistance (megohms/1000 feet, min.)

Conductor to conductor — See table II

Conductor continuity — No failure

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Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation), Method 1 only (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Accelerated service – Sizes 23 only (see table II for load current)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure

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Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSFNW	3	2500
LSFNW	4 and 9	2000
LSFNW	23	1500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-054)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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